

lab 03

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##Exercício 1

```
library(readr)
library(dplyr)
```

Anexando pacote: 'dplyr'

Os seguintes objetos são mascarados por 'package:stats':

filter, lag

Os seguintes objetos são mascarados por 'package:base':

intersect, setdiff, setequal, union

```
airlines <- read_csv("airlines.csv")
```

Rows: 14 Columns: 2

-- Column specification -----

Delimiter: ","

chr (2): IATA_CODE, AIRLINE

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
airlines
```

```
# A tibble: 14 x 2
  IATA_CODE AIRLINE
  <chr>      <chr>
1 UA        United Air Lines Inc.
2 AA        American Airlines Inc.
3 US        US Airways Inc.
4 F9        Frontier Airlines Inc.
5 B6        JetBlue Airways
6 OO        Skywest Airlines Inc.
7 AS        Alaska Airlines Inc.
8 NK        Spirit Air Lines
9 WN        Southwest Airlines Co.
10 DL       Delta Air Lines Inc.
11 EV       Atlantic Southeast Airlines
12 HA       Hawaiian Airlines Inc.
13 MQ       American Eagle Airlines Inc.
14 VX       Virgin America
```

```
airports <- read_csv("airports.csv", col_types = cols(
  IATA_CODE = col_character(),
  CITY = col_character(),
  STATE = col_character(),
  LATITUDE = col_double(),
  LONGITUDE = col_double()
))
airports
```

```
# A tibble: 322 x 7
  IATA_CODE AIRPORT CITY STATE COUNTRY LATITUDE LONGITUDE
  <chr>      <chr> <chr> <chr> <chr> <dbl> <dbl>
1 ABE      Lehigh Valley International~ Alle~ PA USA 40.7 -
75.4
2 ABI      Abilene Regional Airport Abil~ TX USA 32.4 -
99.7
3 ABQ      Albuquerque International S~ Albu~ NM USA 35.0 -
107.
4 ABR      Aberdeen Regional Airport Aber~ SD USA 45.4 -
98.4
5 ABY      Southwest Georgia Regional ~ Alba~ GA USA 31.5 -
84.2
6 ACK      Nantucket Memorial Airport Nant~ MA USA 41.3 -
70.1
```

```

7 ACT      Waco Regional Airport      Waco TX USA      31.6      -
97.2
8 ACV      Arcata Airport              Arca~ CA USA      41.0      -
124.
9 ACY      Atlantic City International~ Atla~ NJ USA      39.5      -
74.6
10 ADK     Adak Airport                Adak AK USA      51.9      -
177.
# i 312 more rows

```

```

limpeza <- function(chunk, pos) {
  chunk %>%
    filter(!startsWith(DESTINATION_AIRPORT, "1")) %>%
    na.omit() %>%
    group_by(DESTINATION_AIRPORT) %>%
    summarise(
      soma_atraso = sum(ARRIVAL_DELAY),
      total_voos = n(),
      .groups = "drop"
    )
}

stats <- read_csv_chunked(
  "flights.csv",
  callback = DataFrameCallback$new(limpeza),
  chunk_size = 1e6,
  col_types = cols_only(
    DESTINATION_AIRPORT = col_character(),
    ARRIVAL_DELAY = col_double()
  )
)

flights <- stats %>%
  group_by(DESTINATION_AIRPORT) %>%
  summarise(
    MEDIA_ATRASO = sum(soma_atraso) / sum(total_voos),
    TOTAL_VOOS = sum(total_voos)
  )

flights

```

```
# A tibble: 322 x 3
```

```

  DESTINATION_AIRPORT MEDIA_ATRASO TOTAL_VOOS
    <chr>              <dbl>         <int>
1 ABE                5.80           2220
2 ABI                4.08           2224
3 ABQ                5.61          18953
4 ABR               -3.39            657
5 ABY                8.68            864
6 ACK                4.91            487
7 ACT               10.6           1541
8 ACV                9.68           1267
9 ACY               11.8           3530
10 ADK               -3.19            89
# i 312 more rows

```

##Exercício 2

Na tabela flights a chave é DESTINATION_AIRPORT. Já na airports é IATA_CODE.

```

ausentes <- flights %>%
  anti_join(airports, by = c("DESTINATION_AIRPORT" = "IATA_CODE"))
print(ausentes)

```

```

# A tibble: 0 x 3
# i 3 variables: DESTINATION_AIRPORT <chr>, MEDIA_ATRASO <dbl>,
#   TOTAL_VOOS <int>

```

```

flights <- flights %>%
  left_join(
    airports,
    by = c("DESTINATION_AIRPORT" = "IATA_CODE")
  )
flights

```

```

# A tibble: 322 x 9
  DESTINATION_AIRPORT MEDIA_ATRASO TOTAL_VOOS AIRPORT      CITY  STATE COUNTRY
  <chr>              <dbl>         <int> <chr>         <chr> <chr> <chr>
1 ABE                5.80           2220 Lehigh Valle~ Alle~ PA    USA
2 ABI                4.08           2224 Abilene Regi~ Abil~ TX    USA
3 ABQ                5.61          18953 Albuquerque ~ Albu~ NM    USA
4 ABR               -3.39            657 Aberdeen Reg~ Aber~ SD    USA
5 ABY                8.68            864 Southwest Ge~ Alba~ GA    USA

```

```

6 ACK          4.91      487 Nantucket Me~ Nant~ MA    USA
7 ACT          10.6      1541 Waco Regiona~ Waco  TX    USA
8 ACV          9.68      1267 Arcata Airpo~ Arca~ CA    USA
9 ACY          11.8      3530 Atlantic Cit~ Atla~ NJ    USA
10 ADK         -3.19       89 Adak Airport  Adak  AK    USA
# i 312 more rows
# i 2 more variables: LATITUDE <dbl>, LONGITUDE <dbl>

```

##Exercício 3

```

aeroportos_por_estado <- airports %>%
  group_by(STATE) %>%
  summarise(QTD_AEROPORTOS = n_distinct(IATA_CODE)) %>%
  arrange(desc(QTD_AEROPORTOS))

aeroportos_por_estado

```

```

# A tibble: 54 x 2
  STATE QTD_AEROPORTOS
  <chr>      <int>
1 TX          24
2 CA          22
3 AK          19
4 FL          17
5 MI          15
6 NY          14
7 CO          10
8 MN           8
9 MT           8
10 NC          8
# i 44 more rows

```

##Exercício 4

```

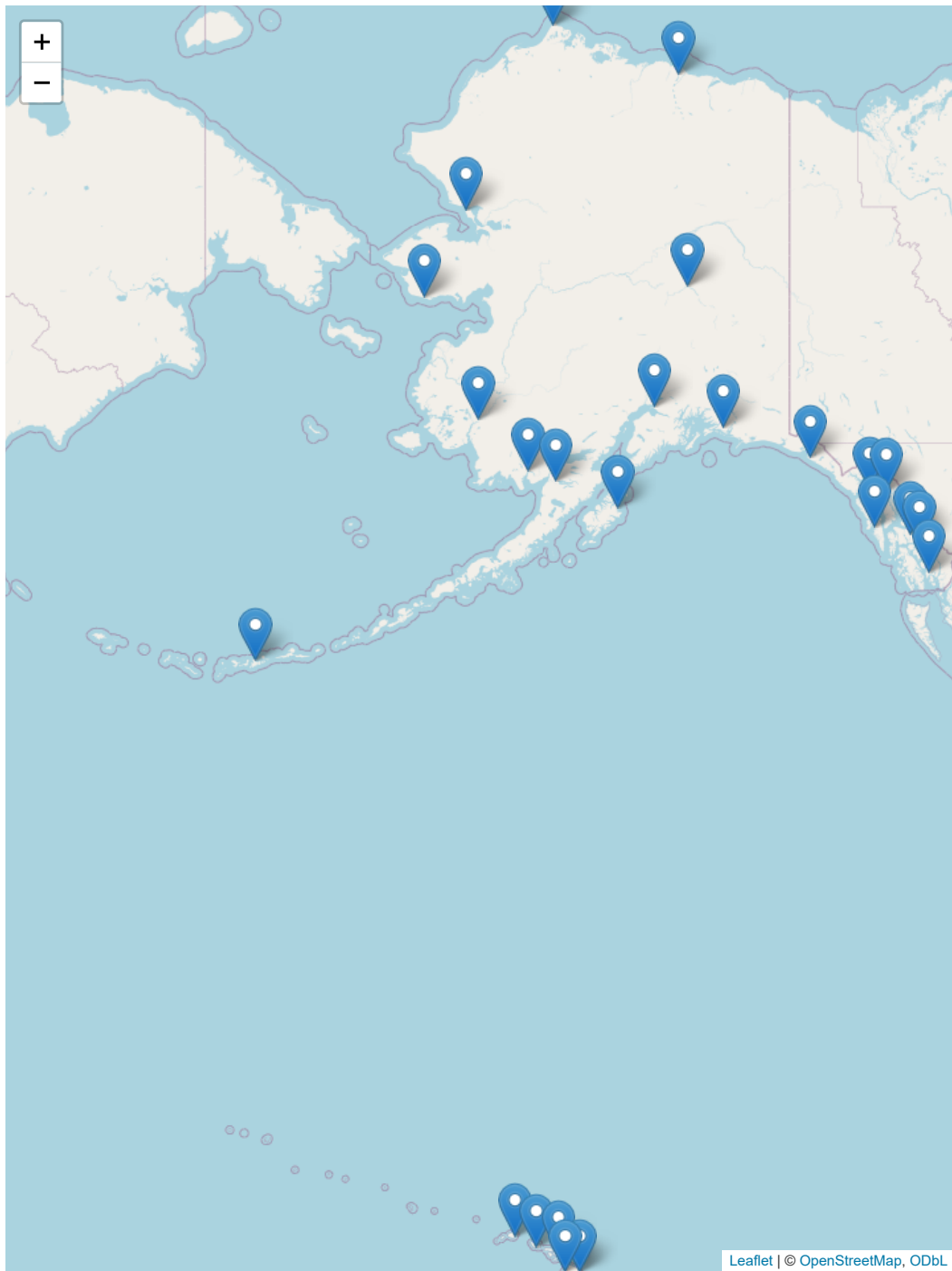
library(leaflet)

this <- flights %>%
  filter(!is.na(LATITUDE), !is.na(LONGITUDE)) %>%
  leaflet() %>%
  addTiles() %>%
  addMarkers(

```

```
lng = ~LONGITUDE,  
lat = ~LATITUDE,  
popup = ~paste0(  
  "<b>Cidade: </b>", CITY, " - ", STATE, "<br>",  
  "<b>Aeroporto: </b>", DESTINATION_AIRPORT, "<br>",  
  "<b>Atraso Médio: </b>", round(MEDIA_ATRASO, 2), " min"  
)  
)  
  
this
```

Warning in file.info(file): cannot resolve owner of file
'\\smb\ra193739\Downloads\ME315\Aula
03\file3958179f31ea\webshot3958563c70.pdf': Não foi feito mapeamento entre os
nomes de conta e as identificações de segurança



```
Sys.setenv(TZ = "America/Sao_Paulo")  
Sys.time()
```

```
[1] "2026-08-31 09:50:45 -03"
```